# Midterms 1, 10-11 | Formulas & graph rules

Growing reference: Midterms 1, 10-11 only. 3 of 28 modules covered. 11 Nov 2026.

## 1. Prices, inflation and real output

$\pi_t = (P_t/P_{t-1} - 1) \times 100\%$

P = price index; π = inflation. Divide the change by the earlier level.

Nominal GDP = $\Sigma\, p_t q_t$; real GDP = $\Sigma\, p_{base} q_t$

Same current quantities. Use current prices for nominal, base-year prices for real (base-year dollars).

Fixed-basket index = $(cost_t/cost_{base}) \times 100$

Keep basket quantities fixed. Inflation uses the index change relative to its earlier level.

GDP deflator = (nominal GDP / real GDP) × 100

Real GDP = nominal GDP ÷ (deflator/100). CPI uses a consumer basket.

$g_{real} = (1+g_{nominal})/(1+\pi_{GDP}) - 1$

Decimal rates. Approximation: real growth ≈ nominal growth − GDP inflation.

Module 1: SRC-011 slides 52-57; SRC-129 p.2; SRC-079 pp.1,3-5.

## 2. Interest and purchasing power

Purchasing quantity = dollars / price per good

$100 ÷ $2 per good = 50 goods. Keep dollars and goods distinct.

$r^e \approx i - \pi^e$; $r^a \approx i - \pi^a$

i = nominal rate; e = expected; a = actual. π (pi) = inflation.

$1+r^a = (1+i)/(1+\pi^a)$

Exact realized return; use decimal rates. Subtraction is an approximation.

Fixed-dollar loan: actual π > expected π

Lower realized real repayment: borrower gains, lender loses relative to expectation.

Module 1: SRC-011 slides 59-62; SRC-129 p.2. Exact formula: derived.

## 3. Wage and after-tax reward

$w = W/P$

W = money wage/hour; P = pre-tax goods price. w = goods/hour.

$R = [(1 - t_n)/(1 + t_c)] \times w$

t_n = labor tax; t_c = consumption tax. Rates in decimals; R = after-tax goods/hour.

Labor tax: 10 × .75 = 7.5; sales tax: 10/1.25 = 8

Separate 25% tax examples at fixed pre-tax real wage 10.

SRC-012 slide 26; SRC-083 pp. 2-4. Tax accounting: derived.

## 4. Individual and aggregate labor

$N_s(w) = \Sigma_i\, h_i(w)$

Add each person's offered hours at the same wage.

Equal individual hours: $N_s$ = pop × h

pop = working-age population; h = offered hours/person.

Population ↓ ⇒ aggregate supply ←

Remaining individual schedules may stay unchanged. More attractive leisure can shift both left.

SRC-012 slides 21, 26, 35. Sum notation: generated.

## 5. Read the labor-market graph

Vertical: w = W/P; horizontal: N

N = total labor hours. Demand slopes down; course supply slopes up with lifetime resources fixed.

$N_d(w^*) = N_s(w^*) = N^*$

Stars = equilibrium. Intersection gives w* and N*.

Same w, fewer supplied hours ⇒ supply LEFT

Wage alone moves along supply. A,K fixed ⇒ demand unchanged.

SRC-012 slides 19-29; SRC-083 p. 2.

## 6. Tax and labor-supply shifts

Temporary tax ↑: $N_s$ ←; N* ↓; w* ↑; R ↓

Set lifetime-resource effect aside; A,K and other shifters fixed; no offsetting transfer.

Permanent tax ↑: substitution ←; income →

Lower work reward favors leisure; lower lifetime resources favor work, assuming normal leisure.

Net N*: ↓, unchanged, or ↑

Compare effect strengths. Exact offset: same N*,w*, but R lower.

SRC-012 slides 23, 25-27, 32-34; SRC-131; SRC-083 pp. 2-6.

## 7. Keep the scenarios separate

More attractive leisure ⇒ individual hours ↓

With this change across households, aggregate supply also shifts left.

Supply ← + demand fixed ⇒ N* ↓; w* ↑

For population/leisure examples, reset taxes and lifetime resources; hold A,K fixed.

Supply shifters: PVLR, taxes, population, VL

PVLR = present value of lifetime resources; VL = value of leisure.

SRC-012 slides 21, 26, 35; SRC-083 pp. 7-8, 11.

## 8. Output, potential output and LRAS

$Y = A\, K^{0.3} N^{0.7}$; $Y^* = A\, K^{0.3} (N^*)^{0.7}$

A = productivity; K = real capital; N* = market-clearing hours.

$Y/N = A(K/N)^{0.3}$

Output per labor hour. Fixed A,K: N ↓ gives Y ↓ but Y/N ↑.

LRAS: P vertical; Y horizontal; Y = Y*

Flexible money wages: doubling P and W leaves W/P,N*,Y* fixed.

A,K or N* change ⇒ LRAS can shift

Potential is equilibrium output, not a physical maximum. P alone does not shift LRAS.

Sticky wage can leave actual Y < current Y*

Lower market-clearing N* alone is not cyclical unemployment.

SRC-012 slides 4, 8, 52-59; SRC-129 p.13.